

# OptiFlow Production Architecture & On-Premise Strategy

This document outlines the recommended production-grade technology stack for OptiFlow, specifically designed for seamless switching between Cloud and On-Premise deployments on Linux OS.

## Recommended Full-Stack Architecture

| Tier               | Technology                 | Purpose                                                                                                |
|--------------------|----------------------------|--------------------------------------------------------------------------------------------------------|
| Frontend UI        | React (Vite) + TailwindCSS | High-performance dashboard, fast rendering for large tables, responsive UI.                            |
| Web Server / Proxy | NGINX                      | Serves static frontend build, routes API traffic, handles HTTPS/SSL certificates.                      |
| Backend Framework  | Python (FastAPI)           | Async Python API layer; high performance, automatic OpenAPI documentation, strict Pydantic validation. |
| Database           | PostgreSQL                 | Primary store for SOH, historical sales, store planograms, dispatch logs, and SKU catalog.             |
| ORM & Migrations   | SQLAlchemy + Alembic       | Python object mapping and database schema version control.                                             |
| Async Task Queue   | Celery + Redis             | Background queue for processing heavy allocation algorithms without freezing the UI.                   |
| Packaging & Deploy | Docker + Docker Compose    | Wraps the entire stack into isolated containers for 1-click cloud or on-prem deployment.               |

# On-Premise Linux Server Environment (Ubuntu / RHEL / Rocky Linux)

For on-premise deployment, OptiFlow is designed to run directly on standard enterprise Linux operating systems (e.g., **Ubuntu Server 22.04 LTS**, **Debian 12**, **RHEL 9**, or **Rocky Linux 9**).

## 1. Linux Server Prerequisites

- **Docker Engine & Docker Compose:** The server only requires Docker installed (`docker-ce` and `docker-compose-plugin`). No manual Python, Node.js, or PostgreSQL installations are needed on the host OS.
- **Network & Firewall:** Open inbound port `80` (HTTP) and port `443` (HTTPS) on the Linux firewall (`ufw allow 80,443/tcp` or `firewall-cmd --add-port={80,443}/tcp`).

## 2. Quick Linux Deployment Commands

```
# 1. Clone repository or extract release bundle
git clone https://github.com/devdekodeglobal/optiflow-poc.git /opt/optiflow
cd /opt/optiflow

# 2. Configure Environment variables
cp .env.example .env

# 3. Build and launch all services in detached background mode
docker compose up -d --build
```

## 3. Automatic System Boot & Auto-Restart (`systemd`)

To ensure OptiFlow automatically starts whenever the physical Linux server reboots or recovers from a power outage, create a simple `systemd` service:

`/etc/systemd/system/optiflow.service:`

```
[Unit]
Description=OptiFlow On-Premise Application Stack
After=docker.service
Requires=docker.service

[Service]
Type=oneshot
RemainAfterExit=yes
WorkingDirectory=/opt/optiflow
ExecStart=/usr/bin/docker compose up -d
ExecStop=/usr/bin/docker compose down

[Install]
WantedBy=multi-user.target
```

Enable the service on Linux: `sudo systemctl enable --now optiflow`

#### 4. Data Persistence & Linux Nightly Backups

- **Database Volume:** PostgreSQL stores data securely on the Linux host filesystem at `/var/lib/docker/volumes/optiflow_postgres_data/_data`.
- **Automated Nightly Backup:** Add a simple Linux `cron` job (`/etc/cron.daily/optiflow-backup`) to perform nightly database dumps:

```
#!/bin/bash
BACKUP_DIR="/var/backups/optiflow"
mkdir -p $BACKUP_DIR
docker exec optiflow-postgres-1 pg_dump -U optiflow_user optiflow_db | gzip >
"$BACKUP_DIR/optiflow_$(date +%Y%m%d).sql.gz"
find $BACKUP_DIR -type f -mtime +30 -delete
```

### Cloud vs. On-Premise Deployment Comparison

| Metric           | Cloud Deployment (AWS / GCP)                              | On-Premise Deployment (Linux Server)                 |
|------------------|-----------------------------------------------------------|------------------------------------------------------|
| Operating System | Managed Linux VM (Ubuntu/ Amazon Linux)                   | Enterprise Linux Server (Ubuntu / RHEL)              |
| Network Endpoint | Public HTTPS Domain ( <code>optiflow.company.com</code> ) | Internal LAN IP ( <code>http://192.168.1.50</code> ) |
| Database Host    | Cloud Managed DB (AWS RDS PostgreSQL)                     | Dockerized PostgreSQL on Linux Disk                  |
| Hardware Control | Virtual Cloud vCPU & RAM                                  | Physical Server Hardware in Warehouse                |
| Portability      | Identical Docker Compose Configuration                    | Identical Docker Compose Configuration               |

### Key Architectural Benefits

1. **Zero Software License Fees:** Runs 100% on free, open-source Linux software stack.

- 2. Resilient On-Premise Recovery:** Linux `systemd` auto-restarts services automatically after server power cycles.
- 3. Isolated Host Environment:** Docker isolates all app dependencies from the main Linux host OS.
- 4. Data Integrity & Automated Backups:** PostgreSQL ACID compliance paired with native Linux automated cron backups.